

# Analog IC Design

## Lecture 08 Frequency Response (1)

**Dr. Hesham A. Omran**

Integrated Circuits Laboratory (ICL)  
Electronics and Communications Eng. Dept.  
Faculty of Engineering  
Ain Shams University

# Outline

- ❑ Recapping previous key results
- ❑ Bode plot review
- ❑ Where are the capacitors?
- ❑ Approximate analysis techniques
  - Short-circuit time constant (SCTC) and open-circuit time constant (OCTC) techniques
  - Dominant pole approximation
- ❑ IC amplifier frequency response
  - Unity gain frequency (UGF) and gain-bandwidth product (GBW)
- ❑ Calculating zeros and poles by inspection
  - Associating poles with nodes
- ❑ Miller's theorem

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# MOSFET in Saturation

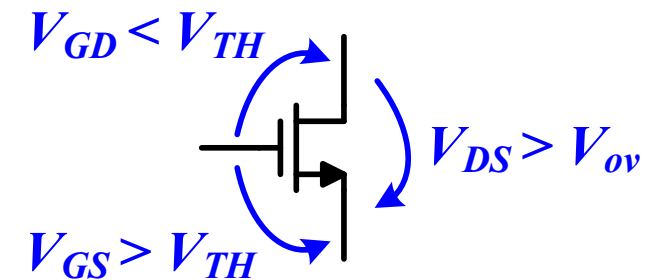
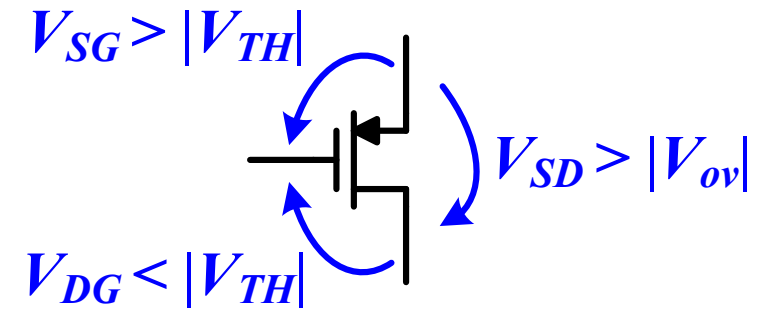
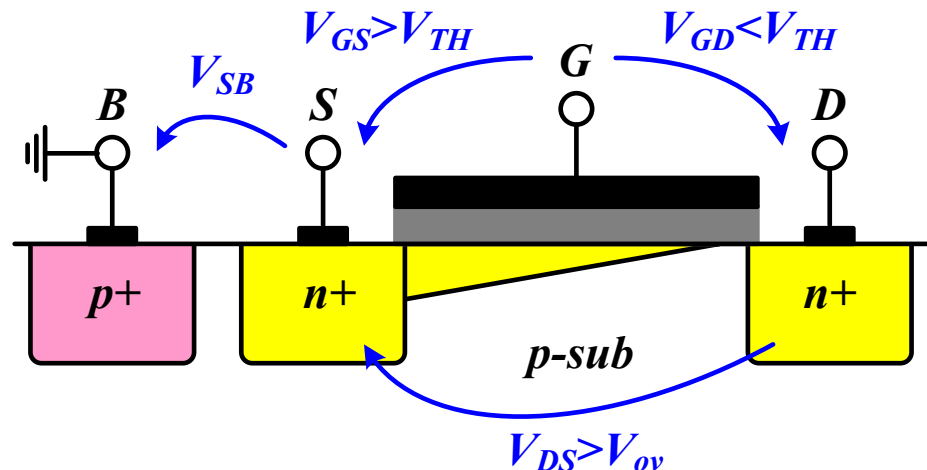
- ❑ The channel is pinched off if the difference between the gate and drain voltages is not sufficient to create an inversion layer

$$V_{GD} \leq V_{TH} \quad \text{OR} \quad V_{DS} \geq V_{ov}$$

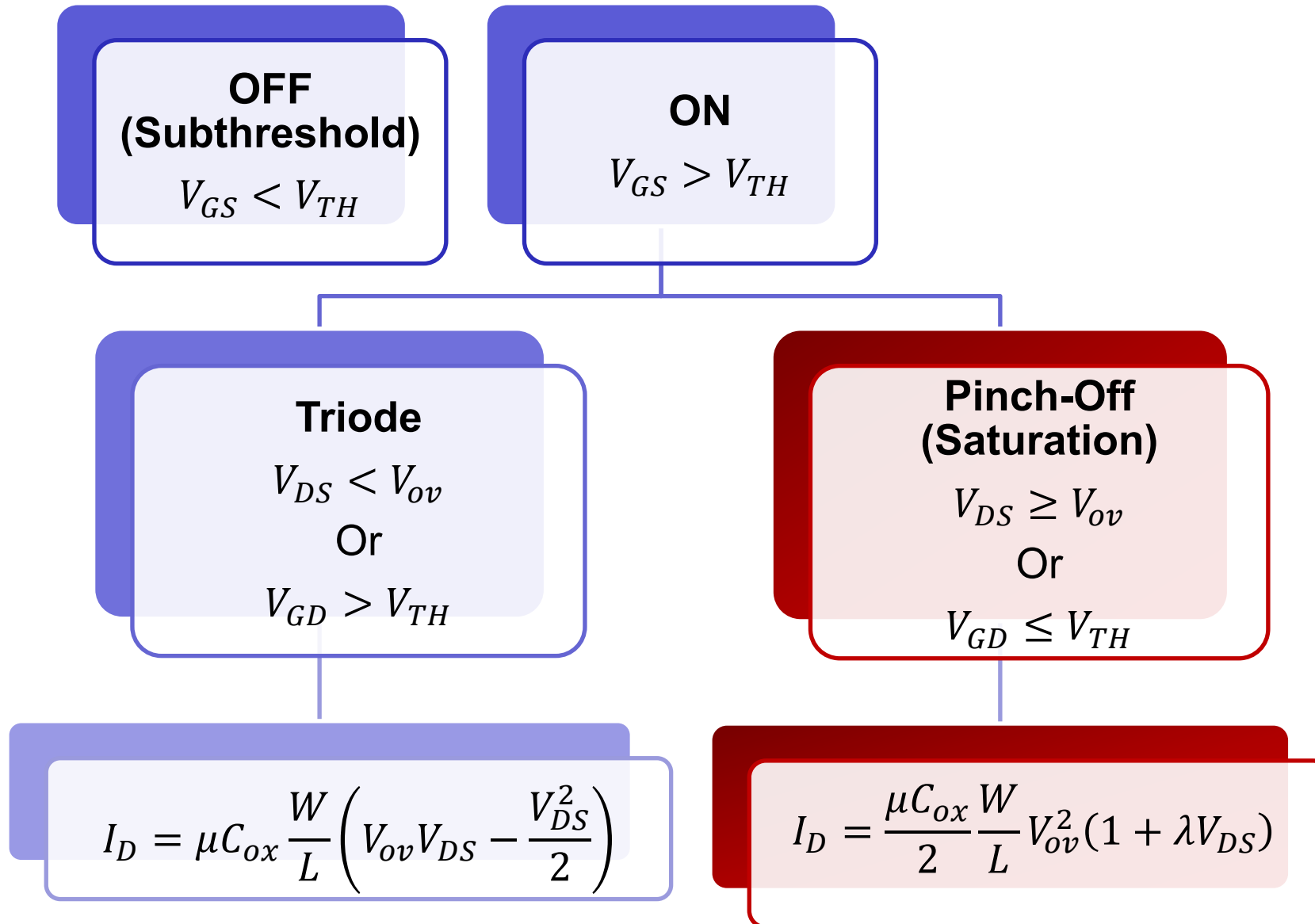
- ❑ Square-law (long channel MOS)

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} \cdot V_{ov}^2 (1 + \lambda V_{DS})$$

$$V_{SB} \uparrow \Rightarrow V_{TH} \uparrow$$



# Regions of Operation Summary



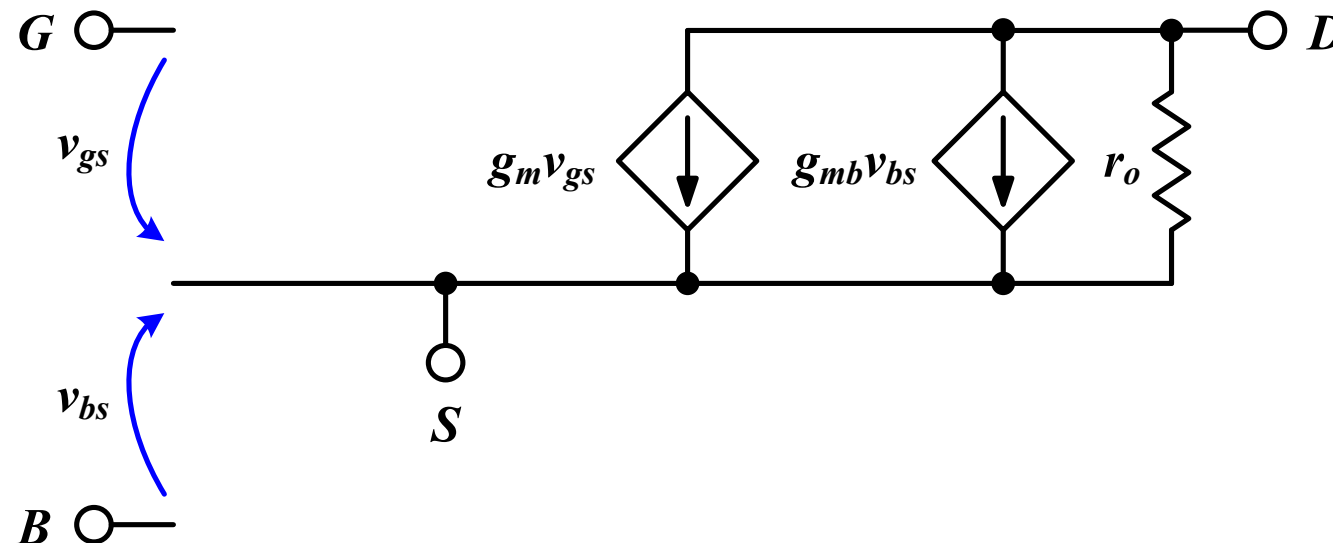
# Low-Frequency Small-Signal Model

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu C_{ox} \frac{W}{L} V_{ov} = \sqrt{\mu C_{ox} \frac{W}{L} \cdot 2I_D} = \frac{2I_D}{V_{ov}}$$
$$g_{mb} = \eta g_m \quad \eta \approx 0.1 - 0.25$$

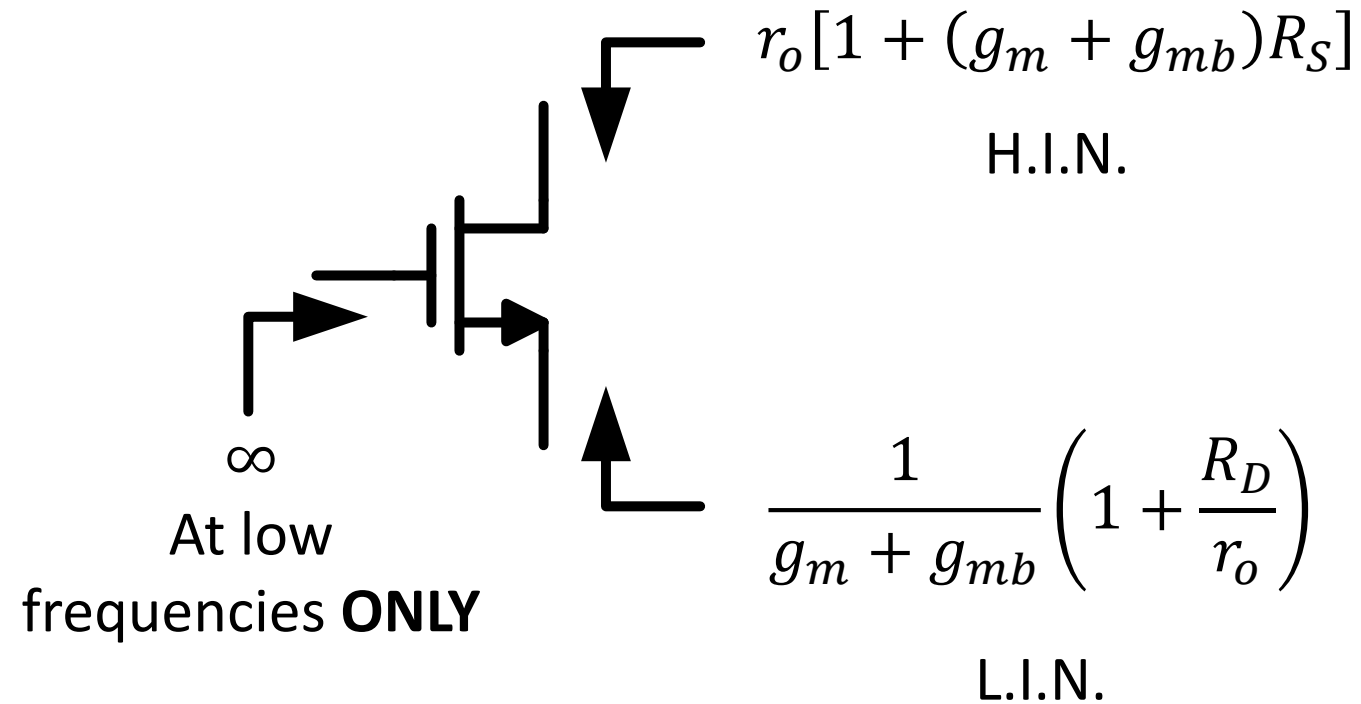
$$r_o = \frac{1}{\partial I_D / \partial V_{DS}} = \frac{V_A}{I_D} = \frac{1}{\lambda I_D}$$

$$V_A \propto L \leftrightarrow \lambda \propto \frac{1}{L}$$

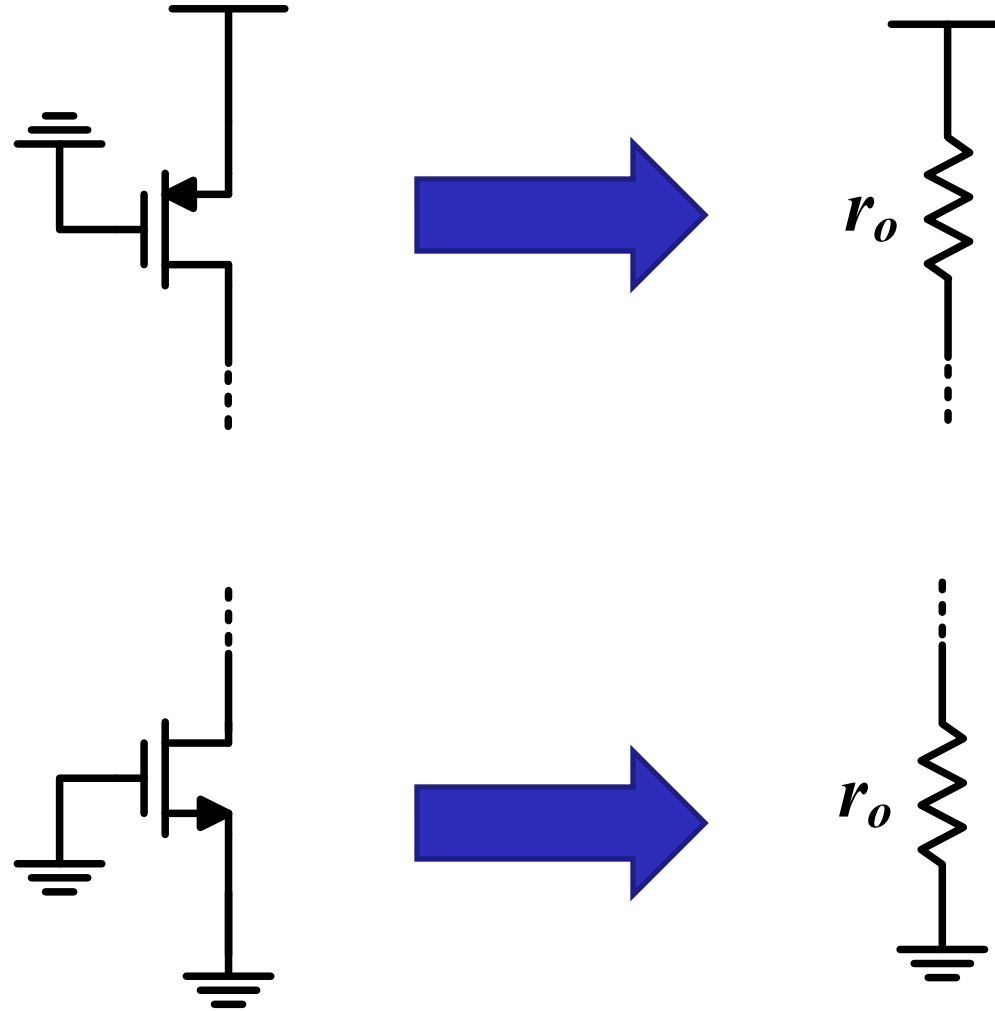
$$V_{DS} \uparrow \quad V_A \uparrow$$



# Rin/out Shortcuts Summary



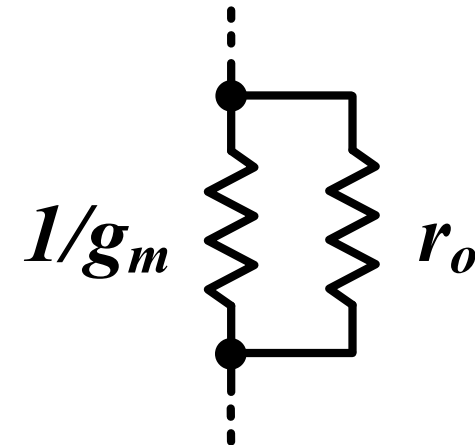
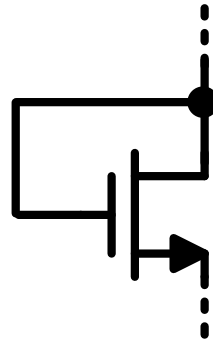
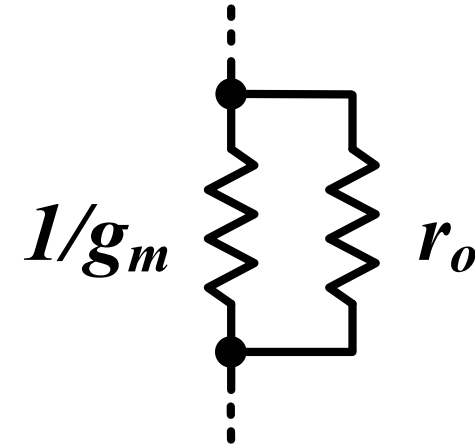
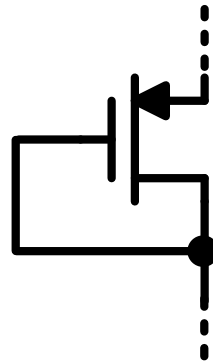
# Active Load (Source OFF)





# Diode Connected (Source Absorption)

- ❑ Always in saturation ( $V_{DS} = V_{GS} > V_{ov}$ )
- ❑ Body effect:  $g_m \rightarrow g_m + g_{mb}$  (if G is ac gnd)



# Why GmRout?

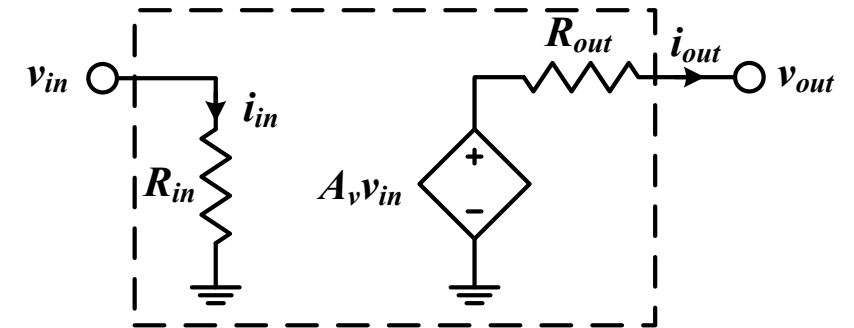
$$R_{in} = v_{in}/i_{in}$$

$$R_{out} = v_x/i_x @ v_{in} = 0$$

$$G_m = i_{out,sc}/v_{in}$$

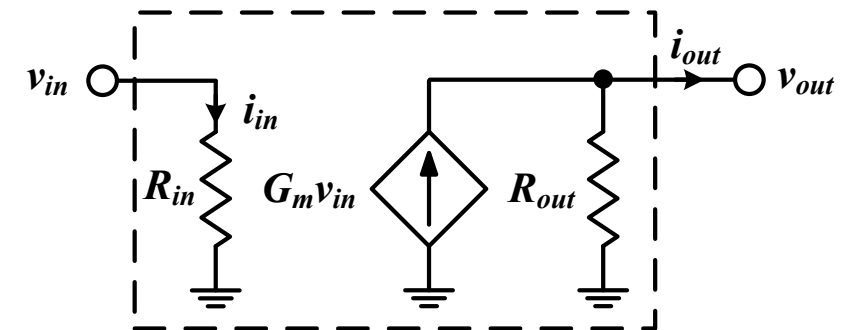
$$A_{v,oc} = G_m R_{out}$$

$$A_{i,sc} = G_m R_{in}$$

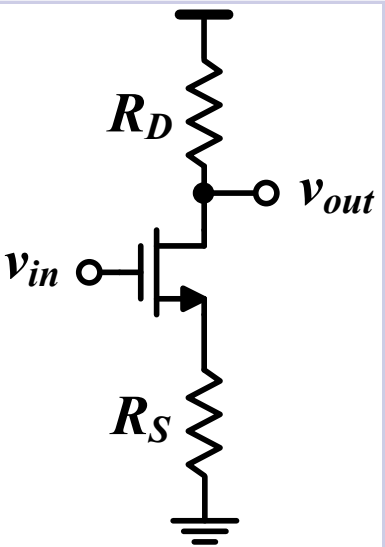
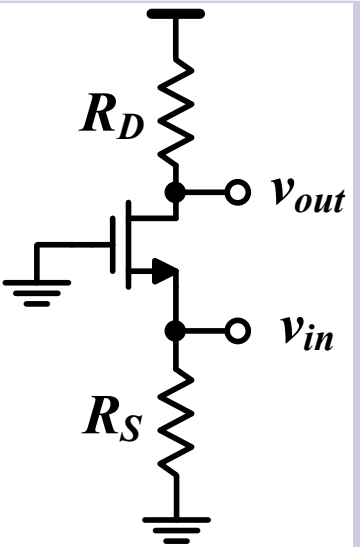
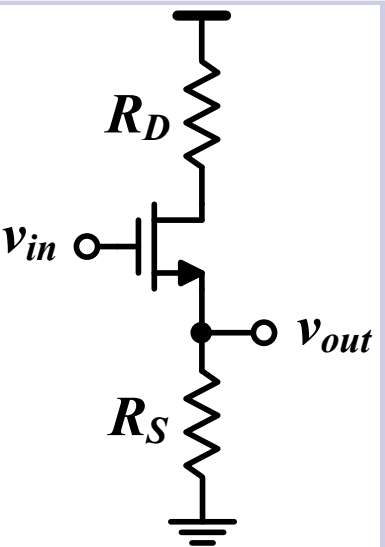


## □ Divide and conquer

- Rout simplified:  $v_{in}=0$
- Gm simplified:  $v_{out}=0$
- We already need  $R_{in}/out$  and  $G_m$
- We can quickly and easily get  $R_{in}/out$  from the shortcuts



# Summary of Basic Topologies

|                        | CS                                                                                 | CG                                                                                  | CD (SF)                                                                             |
|------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                        |  |  |  |
|                        | Voltage & current amplifier                                                        | Voltage amplifier<br>Current buffer                                                 | Voltage buffer<br>Current amplifier                                                 |
| <b>R<sub>in</sub></b>  | $\infty$                                                                           | $R_S \parallel \frac{1}{g_m + g_{mb}} \left( 1 + \frac{R_D}{r_o} \right)$           | $\infty$                                                                            |
| <b>R<sub>out</sub></b> | $R_D \parallel r_o [1 + (g_m + g_{mb})R_S]$                                        | $R_D \parallel r_o$                                                                 | $R_S \parallel \frac{1}{g_m + g_{mb}} \left( 1 + \frac{R_D}{r_o} \right)$           |
| <b>G<sub>m</sub></b>   | $\frac{-g_m}{1 + (g_m + g_{mb})R_S}$                                               | $g_m + g_{mb}$                                                                      | $\frac{g_m}{1 + R_D/r_o}$                                                           |

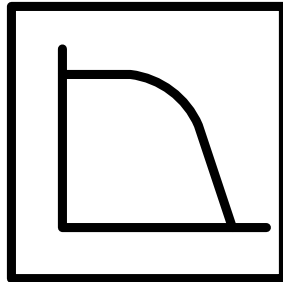
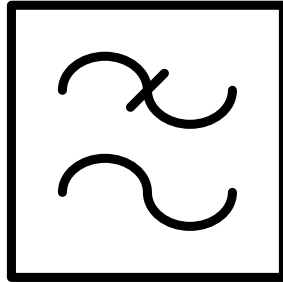
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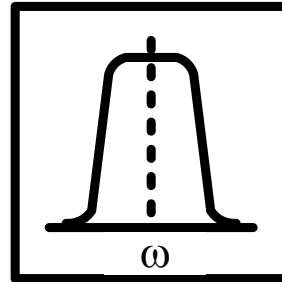
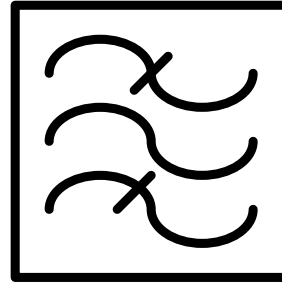
# Frequency Response

□ Y-axis: magnitude of frequency response, x-axis: frequency

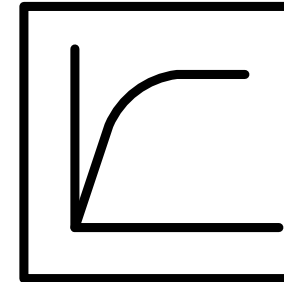
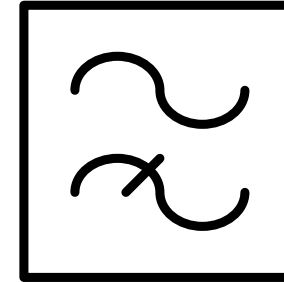
LPF



BPF



HPF



# Poles and Zeros

❑ Transfer function

$$H(s) = \frac{N(s)}{D(s)}$$

❑ Zeros: roots of numerator  $\Rightarrow N(s)$

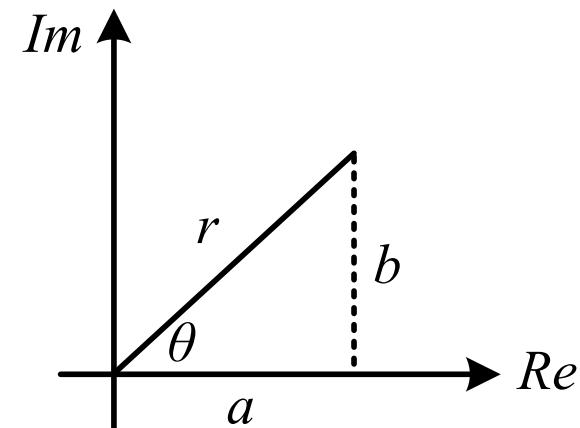
❑ Poles: roots of denominator  $\Rightarrow D(s)$

❑ Frequency response:  $s \Rightarrow j\omega$

$$H(j\omega) = \frac{V_{out}(j\omega)}{V_{in}(j\omega)} = |H(j\omega)|e^{j\phi}$$

❑ Magnitude( $a + jb$ ) =  $r = \sqrt{a^2 + b^2}$

❑ Phase( $a + jb$ ) =  $\theta = \tan^{-1} \frac{b}{a}$



# 1<sup>st</sup> Order LPF

$$H(s) = \frac{v_{out}}{v_{in}} = \frac{1/sC}{R + 1/sC} = \frac{1}{1 + sRC} = \frac{1}{1 + s\tau}$$

$$H(j\omega) = \frac{v_{out}}{v_{in}} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + \frac{j\omega}{\omega_c}}$$

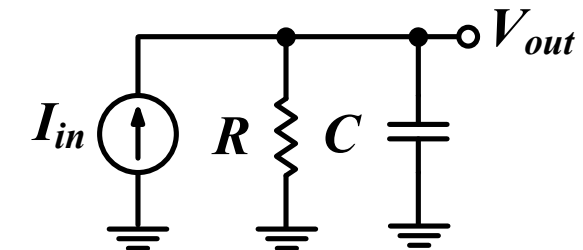
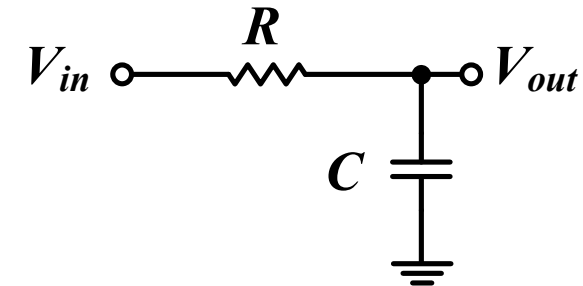
□  $\tau = RC$ : time constant

□  $\omega_c = \frac{1}{\tau} = \frac{1}{RC}$ : cutoff/corner frequency

□ Poles:  $s_p = -\frac{1}{\tau} = -\omega_c$ , Zeros: ?

$$\square |H(j\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}$$

$$\square P(H(j\omega)) = -\tan^{-1} \frac{\omega}{\omega_c}$$



# Bode Plot Rules

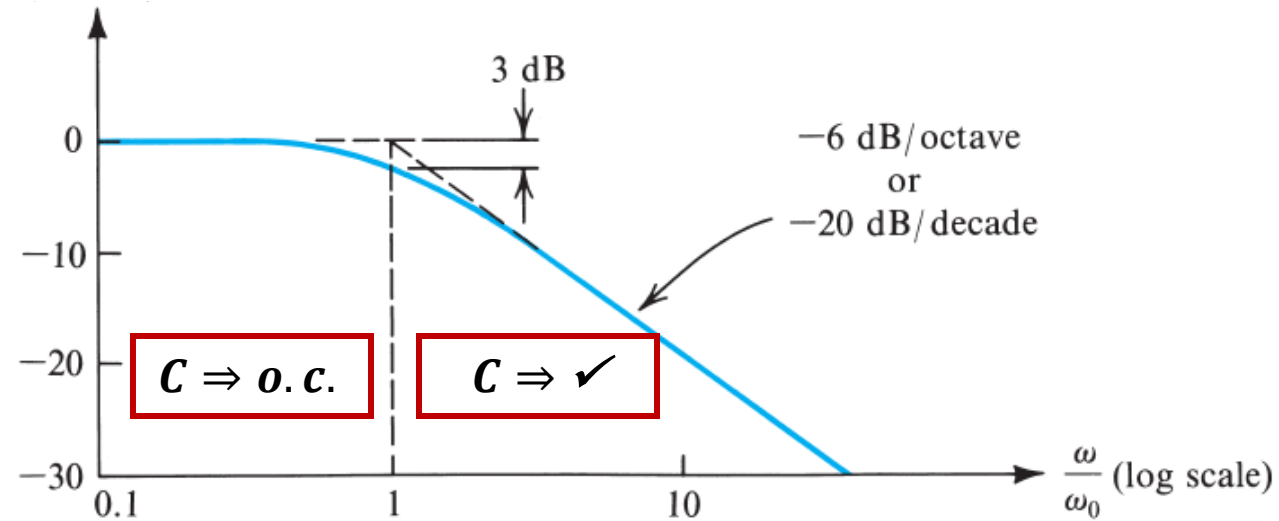
|           | Pole                                      | Zero                                           |
|-----------|-------------------------------------------|------------------------------------------------|
| Magnitude | -20 dB/decade<br>Actual Mag @ pole: -3 dB | +20 dB/decade<br>Actual Mag @ zero: +3 dB      |
| Phase     | -90°<br>Actual Phase @ pole: -45°         | LHP zero:<br>+90°<br>Actual Phase @ zero: +45° |
|           |                                           | RHP zero:<br>-90°<br>Actual Phase @ zero: -45° |

- RHP: Right-half plane ( $\text{Re}\{s\} > 0$ )
- LHP: Left-half plane ( $\text{Re}\{s\} < 0$ )



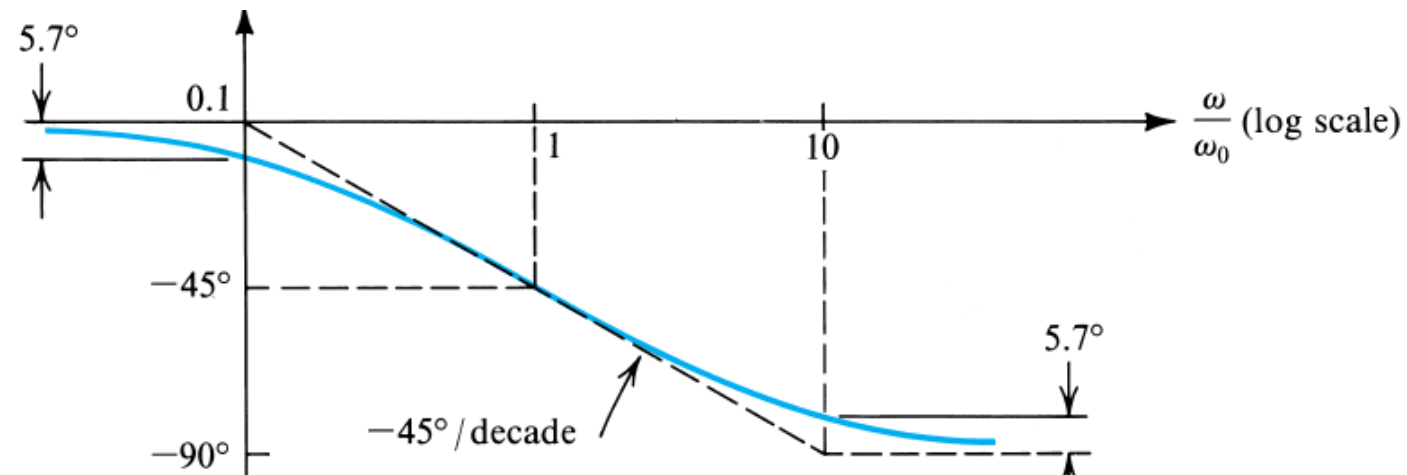
# 1<sup>st</sup> Order LPF Bode Plot

$20 \log|H(j\omega)|$  (dB)



(a)

$P(H(j\omega))$



(b)

# 1<sup>st</sup> Order HPF

$$H(s) = \frac{v_{out}}{v_{in}} = \frac{R}{R + 1/sC} = \frac{sRC}{1 + sRC} = \frac{s\tau}{1 + s\tau}$$

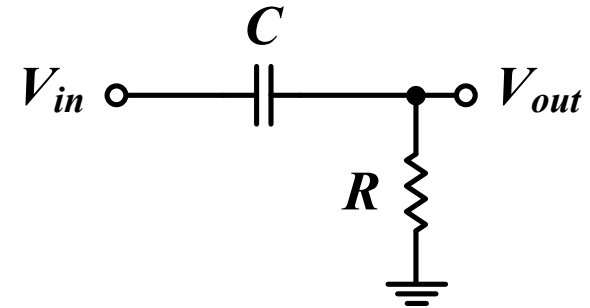
$$H(j\omega) = \frac{v_{out}}{v_{in}} = \frac{R}{R + 1/j\omega C} = \frac{j\omega RC}{1 + j\omega RC} = \frac{\frac{j\omega}{\omega_c}}{1 + \frac{j\omega}{\omega_c}}$$

❑ Poles:  $s_p = -\frac{1}{\tau} = -\omega_c$

❑ Zeros:  $s_z = 0$

❑  $|H(j\omega)| = \frac{\frac{\omega}{\omega_c}}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}$

❑  $P(H(j\omega)) = 90^\circ - \tan^{-1} \frac{\omega}{\omega_c}$



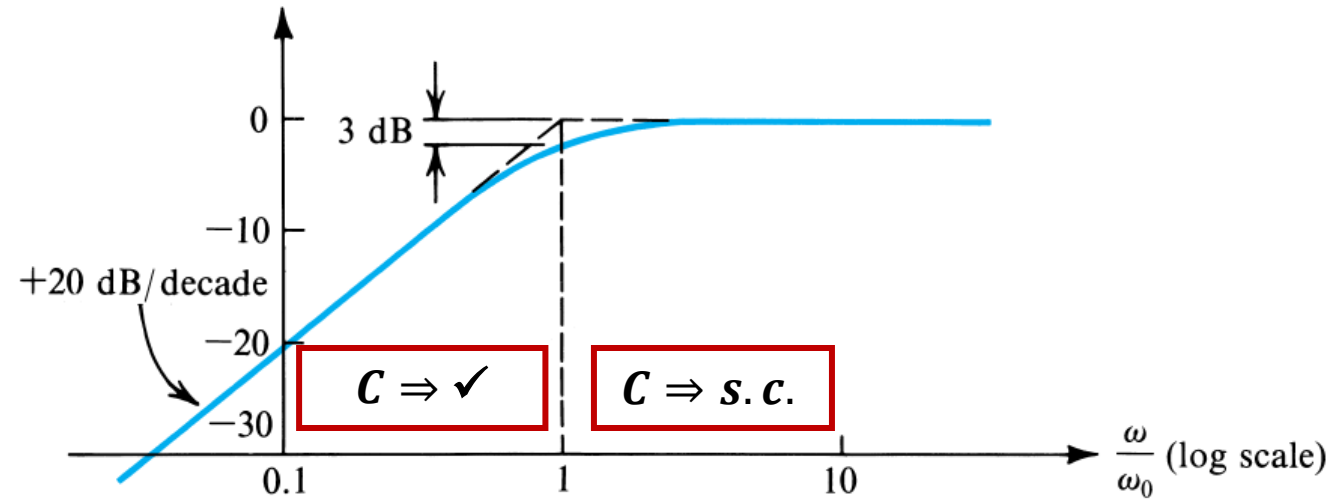
# Bode Plot Rules

|           | Pole                                      | Zero                                           |
|-----------|-------------------------------------------|------------------------------------------------|
| Magnitude | -20 dB/decade<br>Actual Mag @ pole: -3 dB | +20 dB/decade<br>Actual Mag @ zero: +3 dB      |
| Phase     | -90°<br>Actual Phase @ pole: -45°         | LHP zero:<br>+90°<br>Actual Phase @ zero: +45° |
|           |                                           | RHP zero:<br>-90°<br>Actual Phase @ zero: -45° |

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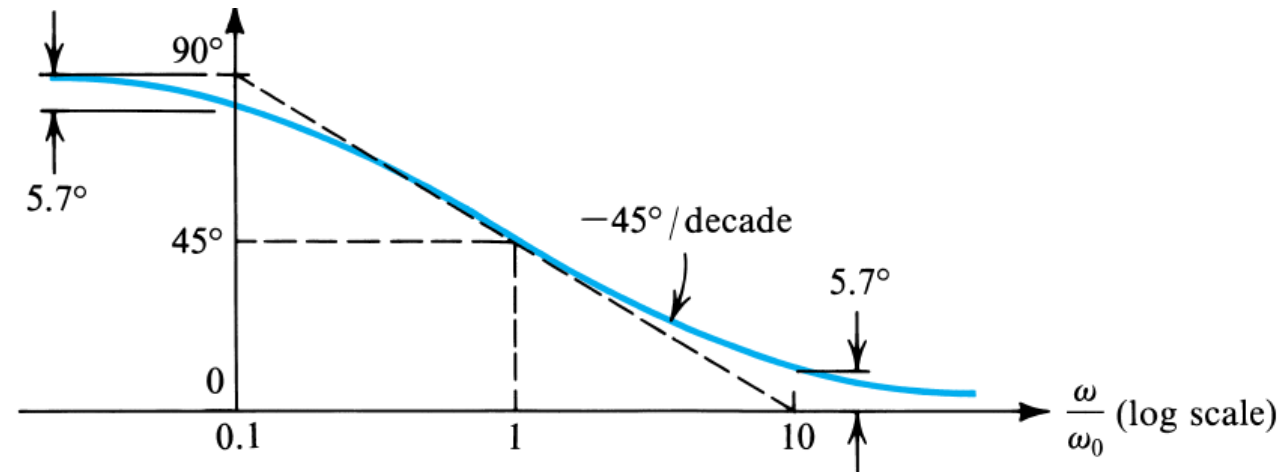
# 1<sup>st</sup> Order HPF Bode Plot

$20 \log|H(j\omega)|$  (dB)



(a)

$P(H(j\omega))$



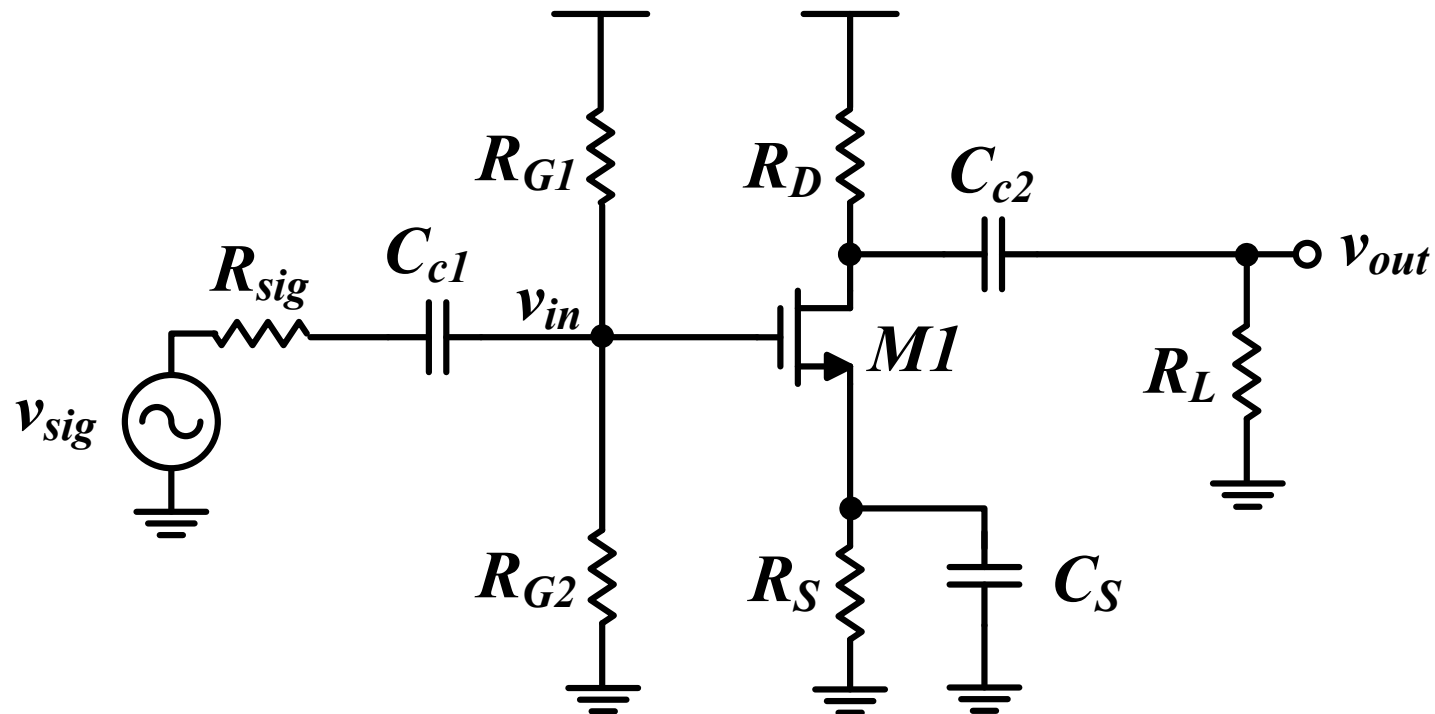
(b)

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# Where are the Capacitors?

- ❑ Coupling capacitors:  $C_{c1}$  and  $C_{c2}$   $\rightarrow$  act as HPF (affect LFR)
- ❑ Bypass capacitor:  $C_S$
- ❑ Usually quite large  $\sim \mu F$



# Effect of Bypass Capacitor

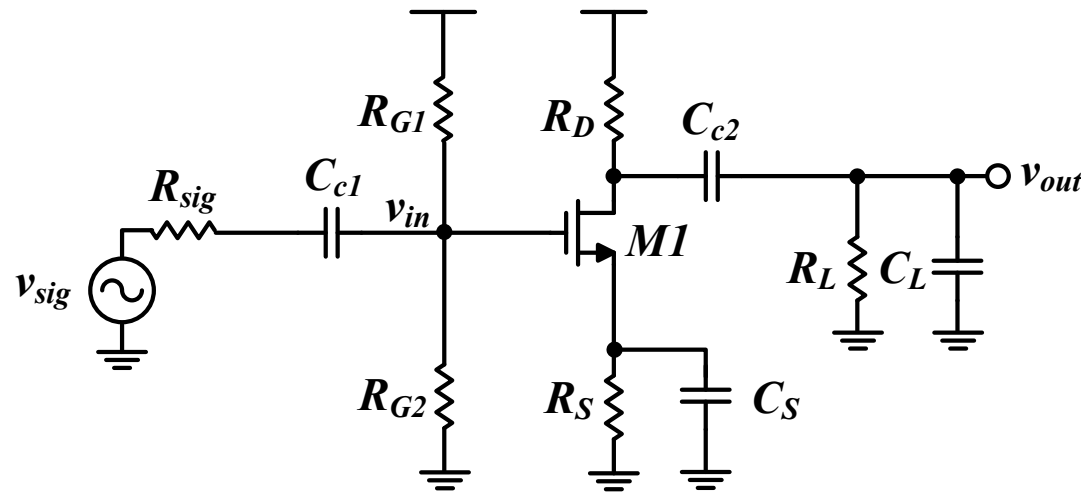
□ Does  $C_S$  act as a LPF or a HPF?

- By intuition, at high frequency  $C_S$  will increase the gain  $\rightarrow$  HPF

$$G_m(s) = \frac{g_m}{1+g_m Z_S} \quad Z_S = \frac{1}{1/R_S + sC_S} = \frac{R_S}{1+sR_SC_S}$$

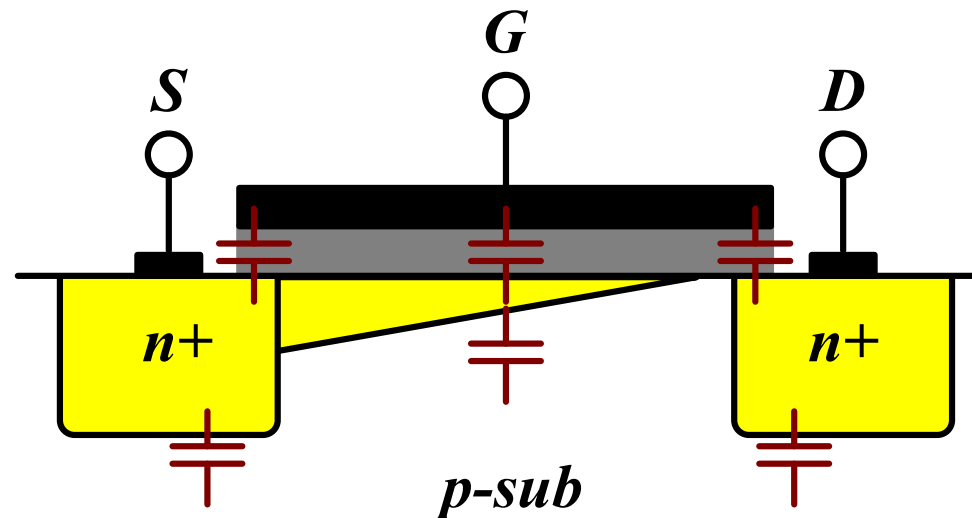
$$G_m(s) = \frac{g_m(1+sR_SC_S)}{(1+g_mR_S)\left(1+\frac{sR_SC_S}{1+g_mR_S}\right)}$$

$$s_z = -\frac{1}{R_SC_S} \text{ and } s_p = -\frac{1+g_mR_S}{R_SC_S} \Rightarrow \omega_p > \omega_z \Rightarrow \text{HPF}$$



# Where are the Capacitors?

- ❑ Gate capacitance ( $C_{gg} = C_{gb} + C_{gs} + C_{gd}$ )
  - Intrinsic part fundamental to MOSFET operation
  - Parasitic part due to the overlap between gate and S/D ( $C_{ov}$ )
- ❑ S/D capacitance ( $C_{sb}$  and  $C_{db}$ )
  - Parasitic capacitances due to reverse biased pn-junctions
    - Bottom-plate ( $C_j$ ) and side-wall components ( $C_{jsw}$ )
- ❑ Usually quite small  $\sim fF$

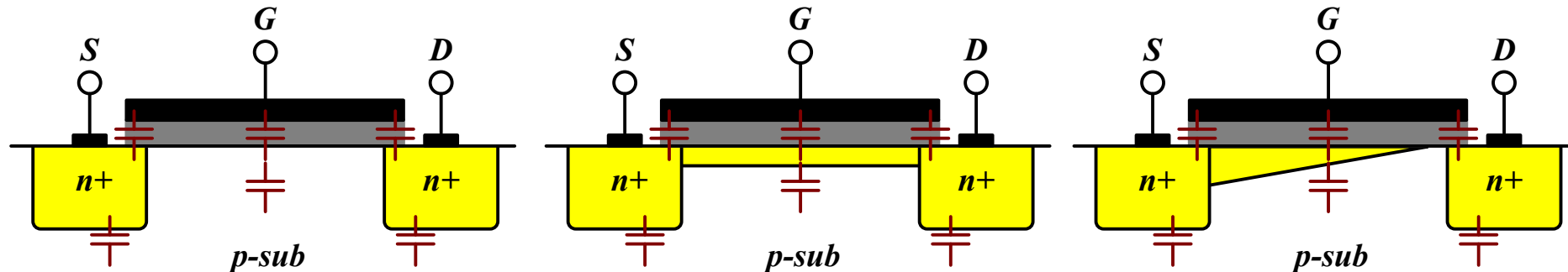




# MOSFET Capacitance

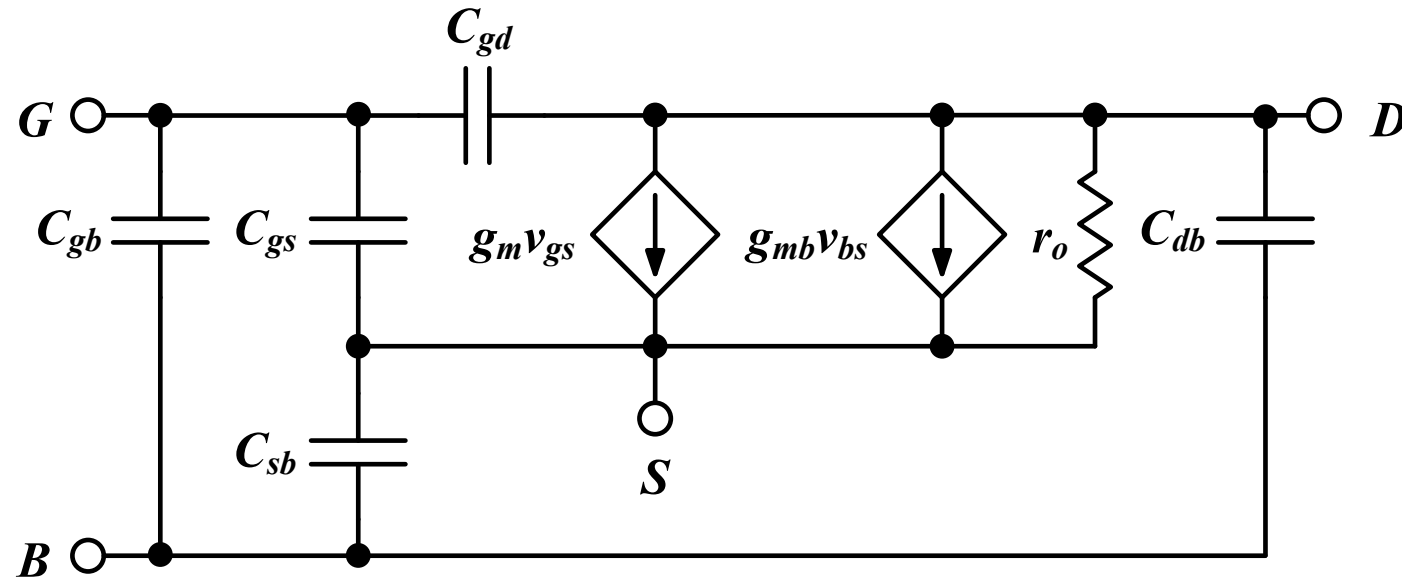
- $C_{ov}$  per unit width,  $C_j$  per unit area, and  $C_{jsw}$  per unit perimeter

|          | Cutoff                  | Deep Triode                                         | Saturation                      |
|----------|-------------------------|-----------------------------------------------------|---------------------------------|
| $C_{gb}$ | $< WLC_{ox}$            | 0                                                   | 0                               |
| $C_{gs}$ | $WC_{ov}$               | $\frac{1}{2}WLC_{ox} + WC_{ov}$                     | $\frac{2}{3}WLC_{ox} + WC_{ov}$ |
| $C_{gd}$ | $WC_{ov}$               | $\frac{1}{2}WLC_{ox} + WC_{ov}$                     | $WC_{ov}$                       |
| $C_{sb}$ | $A_S C_j + P_S C_{jsw}$ | $\left(A_S + \frac{WL}{2}\right) C_j + P_S C_{jsw}$ | $(A_S + WL) C_j + P_S C_{jsw}$  |
| $C_{db}$ | $A_D C_j + P_D C_{jsw}$ | $\left(A_D + \frac{WL}{2}\right) C_j + P_D C_{jsw}$ | $A_D C_j + P_D C_{jsw}$         |



# High Frequency Small Signal Model

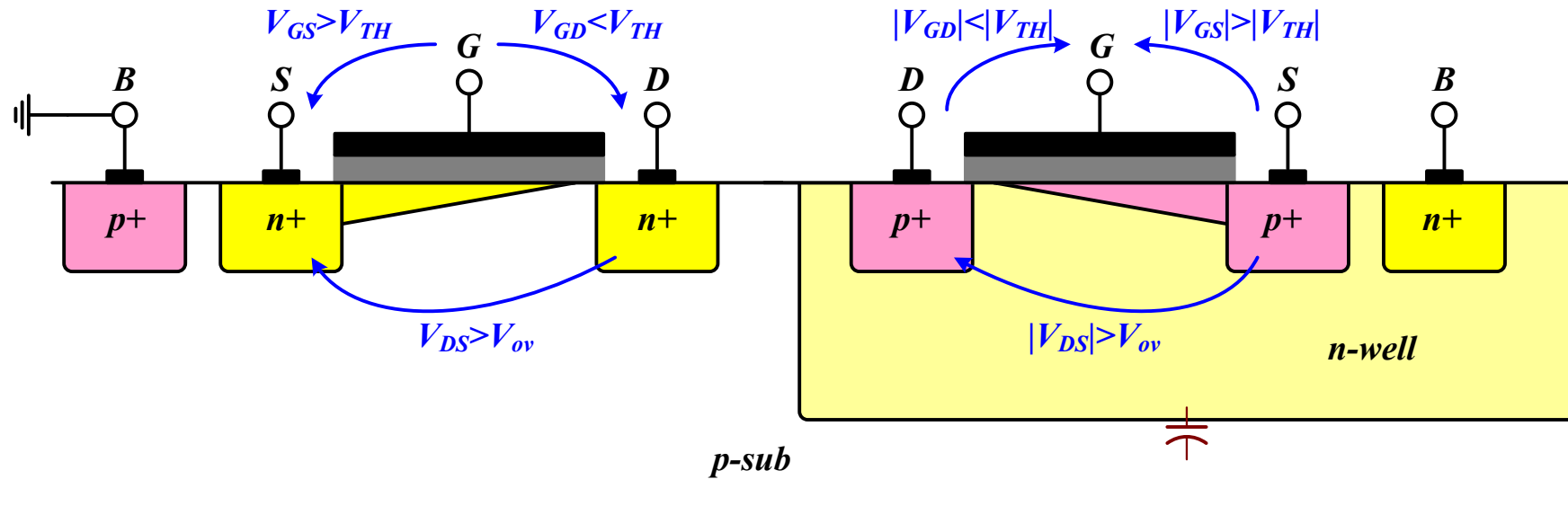
- ❑ MOSFET capacitances act as LPF (affect HFR)
- ❑ In pinch-off saturation
  - $C_{gb} \approx 0$  (for SI only)
  - $C_{gs} \gg C_{gd}$  (not valid for  $L \downarrow\downarrow$ , why?)
  - $C_{sb} > C_{db}$



# N-Well Capacitance



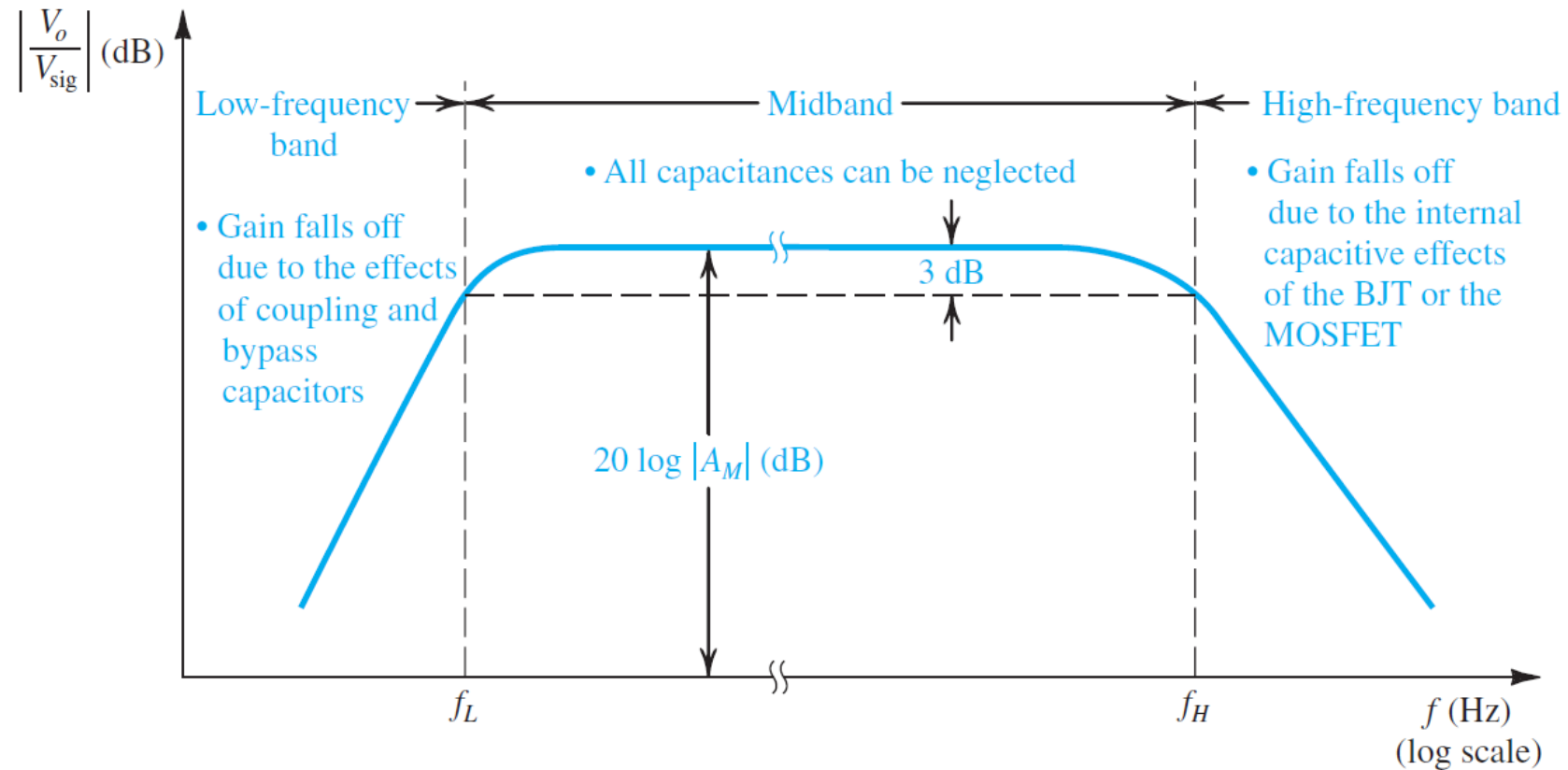
- ❑ There is an additional junction cap between n-well and p-sub
- ❑ If the n-well is tied to VDD this cap is ac shorted (why?)
- ❑ But if n-well is floating (e.g., PMOS S and B connected) this cap is not ac shorted
  - Usually not modeled in SPICE
  - Must be added manually  $\sim 0.05 fF/\mu m^2$



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# Frequency Response



|                         |      |      |      |
|-------------------------|------|------|------|
| Coupling and bypass cap | ✓    | S.C. | S.C. |
| MOSFET and load cap     | O.C. | O.C. | ✓    |

# SCTC and OCTC Techniques

- ❑ Low-frequency range (LFR) => Not common in Analog IC design
  - Only consider one cap at a time → Assume other caps are s.c. → s.c. time constant (SCTC) technique
  - $\omega_{L,3dB} \approx \omega_{L1} + \omega_{L2} + \dots$
  - Highest pole dominates (L.I.N. dominates)
- ❑ **High-frequency range (HFR) => More important in Analog ICs**
  - Only consider one cap at a time → Assume other caps are o.c. → o.c. time constant (OCTC) technique
  - $\omega_{H,3dB} \approx \omega_{H1} || \omega_{H2} || \dots$
  - Lowest pole dominates (H.I.N. dominates)
- ❑ Both SCTC and OCTC provide good approx if one pole is dominant and poles are real

# Dominant Pole Approximation

- Assume the poles are real and widely separated:  $\omega_{p1} \ll \omega_{p2}$

$$\begin{aligned} A_v(s) &\approx \frac{A_o}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)} = \frac{A_o}{1 + \left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}}\right)s + \frac{1}{\omega_{p1}\omega_{p2}}s^2} \\ &\approx \frac{A_o}{1 + \left(\frac{1}{\omega_{p1}}\right)s + \frac{1}{\omega_{p1}\omega_{p2}}s^2} = \frac{A_o}{1 + b_1s + b_2s^2} \\ \omega_{p1} &\approx \frac{1}{b_1} \quad \text{and} \quad \omega_{p2} \approx \frac{1}{b_2\omega_{p1}} = \frac{b_1}{b_2} \end{aligned}$$

- OCTC provides an approx value for dominant pole only
- Dominant pole approx provides an approx value for both dominant and non-dominant poles

# Outline

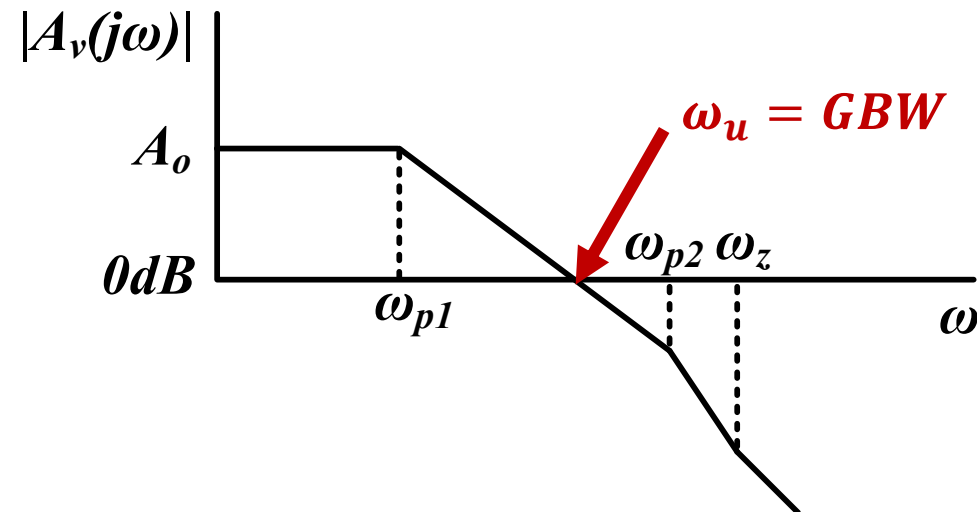
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# IC Amplifier Frequency Response

- ❑  $A_o$  is the low-frequency gain (or DC gain) of the amplifier
- ❑  $\omega_{p1}$  is the dominant pole ( $\omega_{pd}$ )  $\approx$  3dB bandwidth =  $BW = \omega_{3dB}$
- ❑  $\omega_{p2}$  is the non-dominant pole ( $\omega_{pnd}$ )
- ❑ Unity gain frequency (UGF,  $\omega_u$ ) is the frequency at which gain is unity (1 = 0dB)
- ❑ Gain-Bandwidth Product (GBW) =  $Gain \times BW \approx A_o \omega_{p1}$
- ❑ Usually, we design the amplifier such that  $\omega_{p2}$  and  $\omega_z > \omega_u$

$$A_v(s) = \frac{A_o \left(1 + \frac{s}{\omega_z}\right)}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)}$$



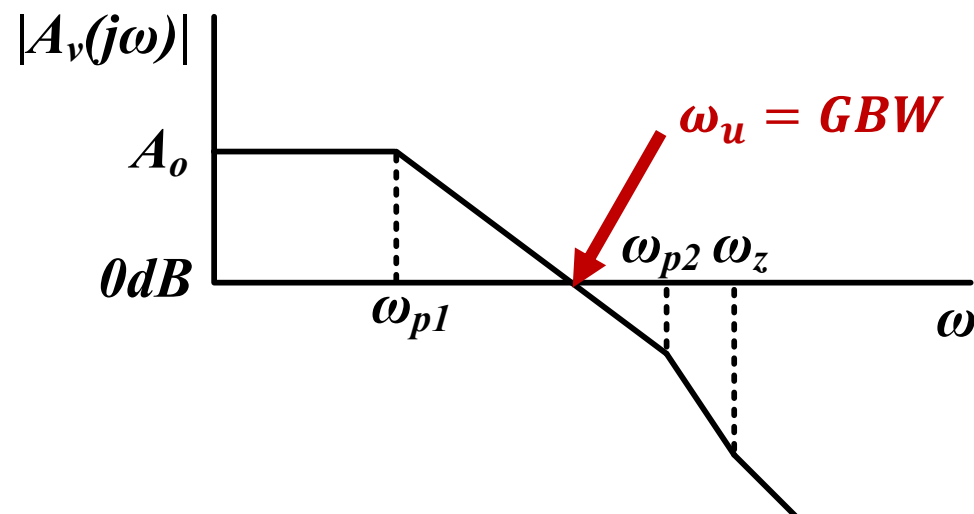
# UGF and GBW

□ At UGF  $\omega = \omega_u$ :  $\omega \gg \omega_{p1}$  and  $\omega \ll \omega_{p2}, \omega_z$

$$|A_v(\omega_u)| = \left| \frac{A_o \left(1 + \frac{j\omega_u}{\omega_z}\right)}{\left(1 + \frac{j\omega_u}{\omega_{p1}}\right) \left(1 + \frac{j\omega_u}{\omega_{p2}}\right)} \right| \approx \left| \frac{A_o}{j\omega_u/\omega_{p1}} \right| = 1 = 0 \text{ dB}$$

$$UGF = \omega_u \approx A_o \omega_{p1} \approx \text{Gain} \times BW = GBW$$

$$A_v(s) = \frac{A_o \left(1 + \frac{s}{\omega_z}\right)}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)}$$



# Outline

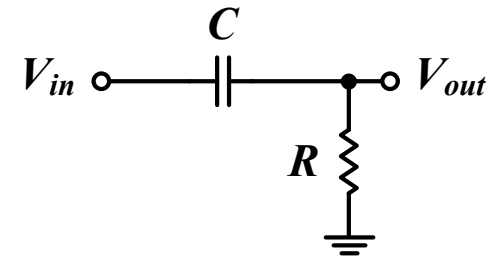
- ❑ Recapping previous key results
- ❑ Bode plot review
- ❑ Where are the capacitors?
- ❑ Approximate analysis techniques
  - Short-circuit time constant (SCTC) and open-circuit time constant (OCTC) techniques
  - Dominant pole approximation
- ❑ IC amplifier frequency response
  - Unity gain frequency (UGF) and gain-bandwidth product (GBW)
- ❑ Calculating zeros and poles by inspection
  - Associating poles with nodes
- ❑ Miller's theorem

# Calculating Zeros by Inspection

1. Find the value  $s = s_z$  that makes  $H(s) = 0 \Rightarrow v_{out} = 0$

□ Ex1:  $C$ :  $v_o = 0$  if  $Z_C = \infty$

- $Z_C = \frac{1}{sC}$
- $\Rightarrow s_z = 0$

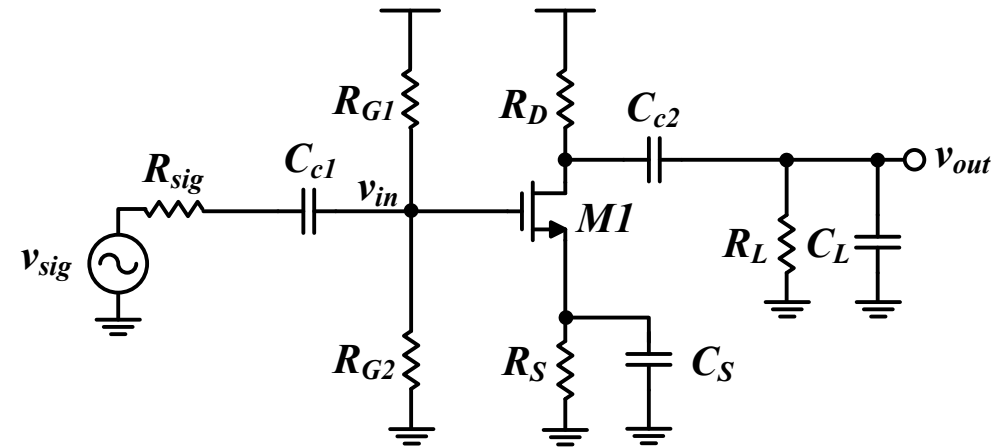


□ Ex2:  $C_{c1}$ :  $v_o = 0$  if  $Z_{C_1} = \infty$

- $Z_{C_1} = \frac{1}{sC_1}$
- $\Rightarrow s_{z1} = 0$

□ Ex3:  $C_S$ :  $v_o = 0$  if  $Z_S = \infty$

- $Z_S = \frac{R_S}{1+sR_SC_S}$
- $\Rightarrow s_{z2} = -\frac{1}{R_SC_S}$



# Calculating Poles by Inspection

1. Set  $v_{sig} = 0$  (deactivate independent sources)
2. Calculate Thevenin resistance ( $R_{th,i}$ ) seen by each cap ( $C_i$ )

$$3. s_{p,i} = -\frac{1}{R_{th,i}C_i}$$

□ Ex1:  $C: R_{th} = R$

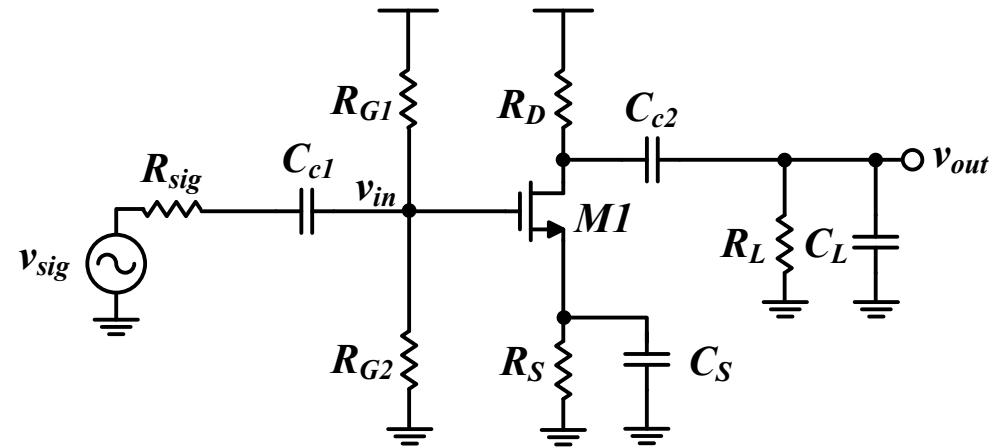
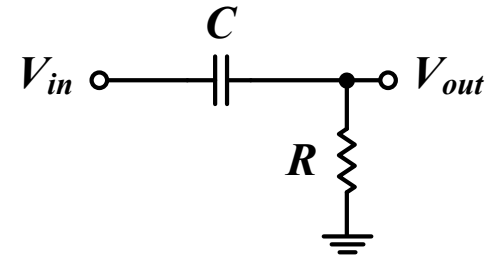
■  $\Rightarrow s_p = -\frac{1}{RC}$

□ Ex2:  $C_{c1}: R_{th} = R_{sig} + R_G$

■  $\Rightarrow s_{p1} = -\frac{1}{(R_{sig} + R_G)C_{c1}}$

□ Ex3:  $C_S: R_{th} \approx R_S \parallel \frac{1}{g_m}$

■  $\Rightarrow s_{p2} = -\frac{1}{\left(R_S \parallel \frac{1}{g_m}\right)C_S}$



# Associating Poles with Nodes

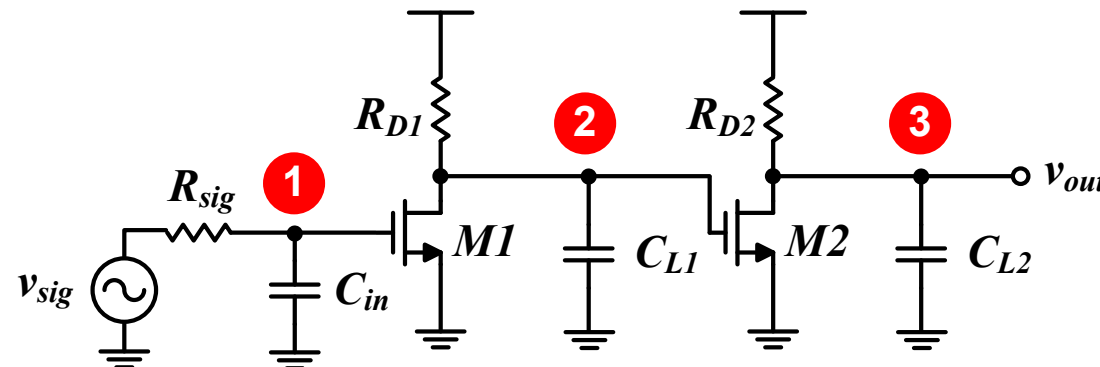
1. Set  $v_{sig} = 0$  (deactivate independent sources)
2. Calculate Thevenin resistance ( $R_{th,i}$ ) seen by each cap ( $C_i$ )

3.  $s_{p,i} = -\frac{1}{R_{th,i}C_i}$

□ Example: Ignore MOSFET  $r_o$  and capacitance

- Each node is associated with a pole
- H.I.N. dominates

$$H(s) = \frac{(g_{m1}R_{D1})(g_{m2}R_{D2})}{(1 + sR_{sig}C_{in})(1 + sR_{D1}C_{L1})(1 + sR_{D2}C_{L2})}$$

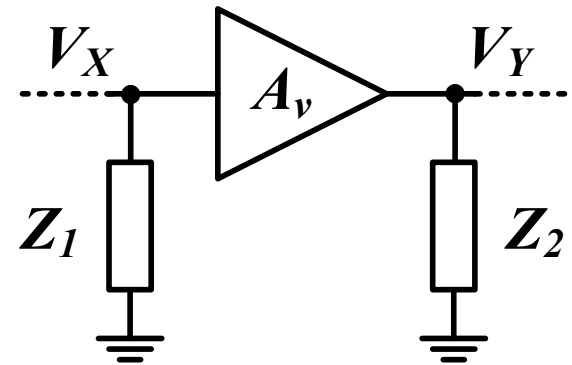
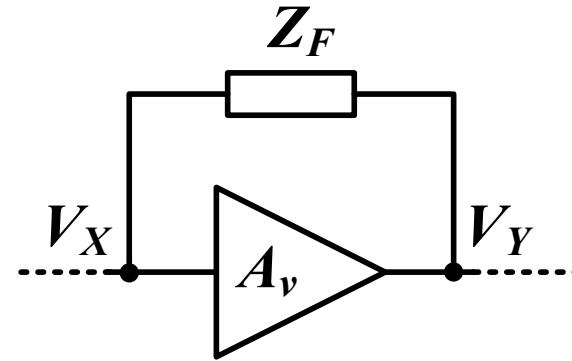


# Outline

- ❑ Recapping previous key results
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# Miller's Theorem

$$A_v = \frac{V_Y}{V_X}$$
$$\frac{V_X - V_Y}{Z_F} = \frac{V_X}{Z_1}$$
$$Z_1 = \frac{Z_F}{1 - \frac{V_Y}{V_X}} = \frac{Z_F}{1 - A_v}$$
$$\frac{V_Y - V_X}{Z_F} = \frac{V_Y}{Z_2}$$
$$Z_2 = \frac{Z_F}{1 - \frac{V_X}{V_Y}} = \frac{Z_F}{1 - \frac{1}{A_v}}$$



❑ Note: Miller's Theorem cannot be used for  $Z_{out}$  calculation (why?)

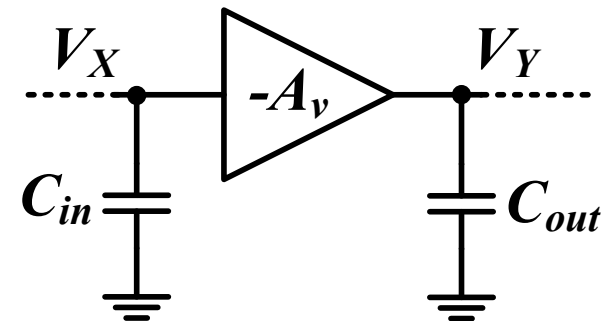
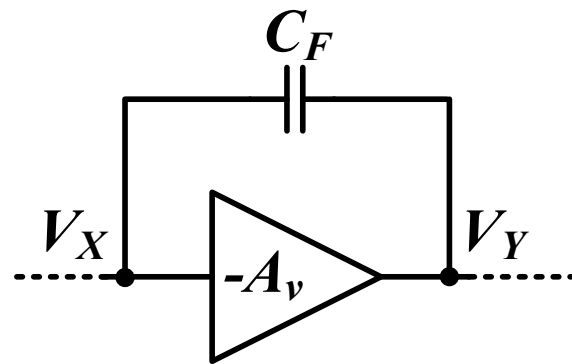


# Miller Effect

□ Miller Effect: Capacitance multiplication if  $Z_F = 1/sC_F$

$$Z_1 = \frac{Z_F}{1 + A_v} \approx \frac{Z_F}{A_v} = \frac{1}{sA_vC_F} \Rightarrow \mathbf{C_{in} = A_vC_F}$$

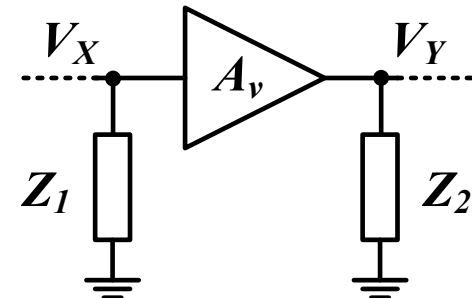
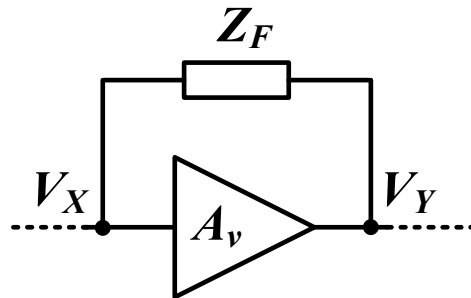
$$Z_2 = \frac{Z_F}{1 + \frac{1}{A_v}} \approx Z_F = \frac{1}{sC_F} \Rightarrow C_{out} \approx C_F$$



# Miller's Approximation

$$Z_1 = \frac{Z_F}{1 - A_v} \quad \& \quad Z_2 = \frac{Z_F}{1 - \frac{1}{A_v}}$$

- ❑ But  $A_v$  is a function of frequency!
- ❑ Miller's Approximation: Substitute with the low frequency gain
  - $A_v(s) \approx A_o$
  - **Gives good approx for the dominant pole ONLY (why?)**
  - It does not tell about the feedforward zero (next slide)



# The Feedforward Zero

□  $v_{out} = 0 \rightarrow i_{out} = 0$

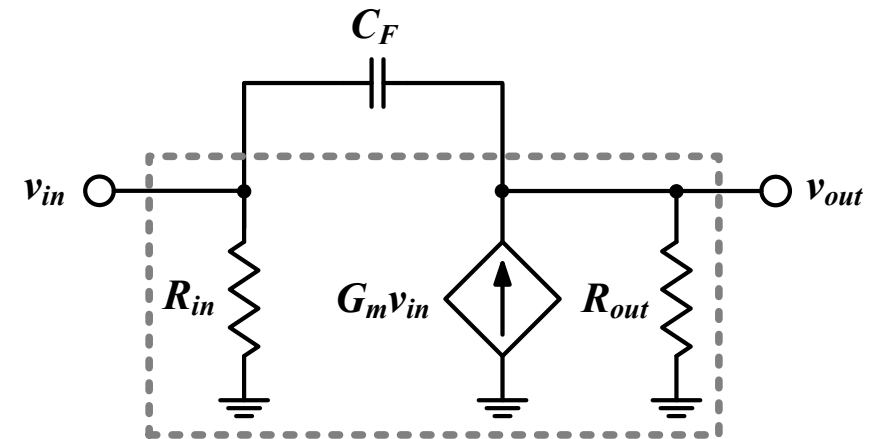
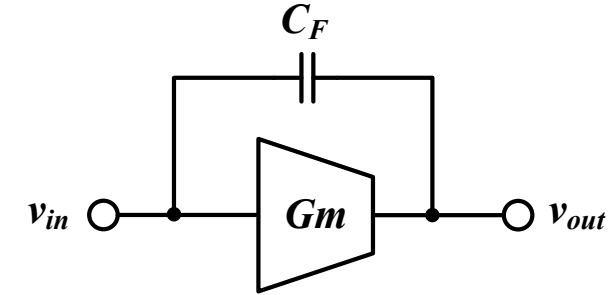
□  $v_{in} s C_F = -G_m v_{in}$

$$s_z = -\frac{G_m}{C_F}$$

□ LHP zero if  $G_m$  is +ve (e.g. CD)

□ **RHP zero if  $G_m$  is -ve (e.g. CS)**

- Mag inc and phase drops
- Very bad for FB loop stability
- More on this when we study op-amp design



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**Thank you!**

# References

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- ❑ N. Weste and D. Harris, “CMOS VLSI Design,” 4<sup>th</sup> ed., Pearson, 2010
- ❑ T. C. Carusone, D. Johns, and K. W. Martin, “Analog Integrated Circuit Design,” 2<sup>nd</sup> ed., Wiley, 2012
- ❑ R. J. Baker, “CMOS circuit design,” 3<sup>rd</sup> ed., Wiley, 2010
- ❑ B. Murmann, EE214 Course Reader, Stanford University